

# Nicolo Foppa Pedretti

+1(917)-294-6771 | [nicolo.fp@gmail.com](mailto:nicolo.fp@gmail.com) | [GitHub: nicolofp](https://github.com/nicolofp) | [LinkedIn](#)

Quantitative Data Scientist with 8 years spanning research, energy, and finance sectors .

Specialized in statistical modeling, machine learning, and Bayesian methods with proven publication track record.

## Experience

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### Praxis Precision Medicine

Feb 2026 – Current

*Associate Director, Statistical Data Scientist*

*Remote, US*

- Build metadata-driven R/Python automation for analytical datasets and TLFs, translating SAP + YAML/CSV specs into reproducible, submission-ready deliverables with full traceability/audit readiness
- Implemented and validated key statistical methods (MMRM, ANCOVA, logistic regression) and built/validated R packages and tooling for exploratory and confirmatory analyses (e.g., mrmr, emmeans)
- Integrated workflows into Databricks + CI/CD, establishing code review, unit testing, and automated validation for continuous reproducibility and quality
- Collaborated cross-functionally (programming, biostatistics, data standards) to align dataset definitions/derivations with controlled standards; mentored team members on best practices

### Critical Path Institute

Sep 2024 – Feb 2026

*Quantitative Medicine Scientist (Data Scientist) – Pharmacometrics*

*Remote, US*

- Built longitudinal Alzheimer's disease progression models (NLME; frequentist/Bayesian) in R/Stan to quantify trajectories and identify predictive biomarkers
- Automated SDTM/ADaM-to-datamart ETL in R with integrated data quality checks and validation

### Icahn School of Medicine at Mount Sinai

Feb 2019 – Apr 2024

*Biostatistician II (Data Scientist) – Environmental Medicine and Climate Science*

*New York, NY*

- Analyzed multi-omics (metabolomics, epigenetics) with ML, DR (PCA/t-SNE/UMAP), and mixed-effects models in R for biomarker and pathway validation
- Developed a Bayesian mixture regression R package (Stan) and applied time-series/exposure models (DLM, ARIMA, splines, SSM) in Stan/JAGS/R
- Built ML pipelines (XGBoost, tree-based, polynomial regression) for biomarker prediction/validation on large-scale genomic data in R/Julia
- Delivered reproducible ETL/workflows (SQL, R, DuckDB) and publication-ready reports (RMarkdown/Quarto; ggplot2/plotly)

### InchCapital

Apr 2017 – Dec 2018

*Quantitative Analyst – R&D*

*Milan, Italy*

- Designed and backtested multi-asset quant strategies using time-series models (ARIMA, HMM, harmonic regression) with Monte Carlo simulations in R/Python/Julia
- Optimized portfolios (Markowitz + advanced methods), defined risk/return KPIs, and automated executive/client reporting in R/RMarkdown

### Gala S.p.A.

Jun 2016 – Nov 2018

*Data Scientist – Market Intelligence and Risk Control*

*Milan, Italy*

- Modeled portfolio performance and commodity price forecasts using statistical/ML methods on energy time series in R
- Validated trading models and managed PostgreSQL ETL; scaled workflows via Linux parallelization and custom RShiny automation (R/SQL)

## Education

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### Università degli Studi Milano-Bicocca

Sep 2013 – Jan 2016

*Master of Science in Statistics*

*Milan, Italy*

### Università degli Studi di Milano

Sep 2009 – Dec 2013

*Bachelor of Science in Applied Mathematics*

*Milan, Italy*

## Skills

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**Languages:** Italian (native), English (professional/advanced)

**Programming Languages:** R, Stan, Julia, Python, SQL, Bash

**Development Frameworks:** RShiny, RMarkdown, Quarto, DuckDB